PH7302: Condensed Matter Physics I

Module 5B: Dielectrics, Ferroelectrics, and Optical Properties of Solids

Comprehensive Student Study Guide & Reference Notes

Department of Physics, National Institute of Technology Calicut

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Target Audience:	M.Sc. Physics / Ph.D. Core Course **Course Code:** PH7302E
Recommended Texts:	C. Kittel, *Introduction to Solid State Physics* (Ch. 14–16); N. W. Ashcroft & N. D. Mermin, *Solid State Physics*; NPTEL Advanced CMP Series.
Module Scope:	Macroscopic electrostatics in matter, local field corrections, Clausius–Mossotti relation, polarizability mechanisms, piezoelectricity, ferroelectricity and domain switching, soft mode theory, complex dielectric response, Lorentz oscillator model, Kramers–Kronig relations, excitons, and Reststrahlen polaritons.

Key Study Takeaway: Curriculum Roadmap & Core Themes

This document serves as the self-contained student companion for Module 5B. It is structured into two core theoretical pillars followed by a unified mathematical summary and comprehensive problem set:

1. **Part I: Dielectric Media & Electromechanical Coupling** — Connecting macroscopic Maxwell fields $\mathbf{E}$ to atomic local fields $\mathbf{E}_{\text{loc}}$ via the Lorentz cavity method; deriving the Clausius–Mossotti relation; analyzing electronic, ionic, and orientational polarizabilities; and establishing the symmetry criteria for piezoelectricity.
2. **Part II: Ferroelectricity, Soft Modes & Optical Response** — Unraveling spontaneous polarization, perovskite crystal chemistry ($BaTiO_3$), domain thermodynamics and P–E hysteresis; connecting structural phase transitions to lattice dynamics via the Lyddane–Sachs–Teller (LST) relation; and exploring optical dispersion, interband transitions, and Reststrahlen phonon-polaritons.
3. **Part III: Summary of Governing Equations** — Complete master formula sheet and diagnostic tables comparing electronic vs. ionic materials.
4. **Part IV: Worked Examination Problems & Conceptual Viva Questions** — Fully worked numerical problems (P1–P7) with complete algebraic derivations and in-depth physical explanations for viva voce oral examinations.

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1 Dielectric Properties of Insulators and Piezoelectricity

1.1 Macroscopic Electrostatics in Insulators

In vacuum, the fundamental electrostatic field is $\mathbf{E}_0$. In an insulating condensed matter system, an applied macroscopic field $\mathbf{E}$ induces bound microscopic charge displacements.

1.1.1 Polarization Vector and Electric Displacement

The electric polarization $\mathbf{P}$ is defined as the net electric dipole moment per unit volume:

$$\mathbf{P} = \lim_{\Delta V \to 0} \frac{1}{\Delta V} \sum_{i \in \Delta V} \mathbf{p}_i = \sum_j N_j \langle \mathbf{p}_j \rangle \tag{1}$$

where N_j is the concentration of polarizable units of type j, and $\langle \mathbf{p}_j \rangle$ is their thermally averaged dipole moment.

In Maxwell's macroscopic equations, the presence of bound charge density $\rho_b = -\nabla \cdot \mathbf{P}$ and free charge density ρ_f is unified through the electric displacement field $\mathbf{D}$:

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} \tag{2}$$

For a linear, isotropic, and homogeneous (LIH) medium:

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E} \tag{3}$$

$$\mathbf{D} = \epsilon_0 (1 + \chi_e) \mathbf{E} = \epsilon_0 \epsilon_r \mathbf{E} = \epsilon \mathbf{E} \tag{4}$$

where χ_e is the electric susceptibility and $\epsilon_r = 1 + \chi_e$ is the relative permittivity (dielectric constant). In anisotropic crystals, χ_e and ϵ_r are second-rank tensors: $D_i = \epsilon_0 \epsilon_{ij} E_j$.

1.1.2 The Dilute-Gas Baseline and Where It Breaks Down

If the polarisable entities are far apart — a dilute gas — the mutual dipole–dipole interaction is negligible, and each atom responds only to the macroscopic field:

$$\mathbf{p} = \alpha \mathbf{E} \tag{5}$$

With the SI convention used throughout these notes (α in units of $\mathrm{F\,m^2}$), the polarization and the susceptibility follow at once:

$$\mathbf{P} = N\mathbf{p} = N\alpha\mathbf{E} \implies \chi_e = \frac{N\alpha}{\epsilon_0} \implies \epsilon_r = 1 + \frac{N\alpha}{\epsilon_0} \tag{6}$$

In Gaussian CGS units the same statement reads $\chi_e = 4\pi N\alpha$ and $\epsilon_r = 1 + 4\pi N\alpha$, with α then measured in $\mathrm{cm^3}$.

This elementary gas law is the baseline against which the whole of Lecture 1 is measured, and it is worth writing on the board first because everything that follows is a correction to it. In a condensed phase the law fails, not because the linear response $\mathbf{p} = \alpha\mathbf{E}$ breaks down, but because the field that acts on a given atom is no longer the macroscopic one: the polarised surroundings generate a **local field** $\mathbf{E}_{\mathrm{loc}}$ that differs from $\mathbf{E}_{\mathrm{macro}}$:

$$\mathbf{E}_{\mathrm{loc}} \neq \mathbf{E}_{\mathrm{macro}} \tag{7}$$

Recovering that correction is the task of the next two subsections.

1.2 The Local Electric Field at an Atomic Site

The field acting directly on an individual atom is **not** the macroscopic Maxwell field $\mathbf{E}$, because $\mathbf{E}$ represents an average over macroscopic volumes containing billions of atoms. The field acting at a specific site $\mathbf{r}_0$ is the **local field** $\mathbf{E}_{\text{loc}}$.

1.2.1 Lorentz Cavity Construction

Following H. A. Lorentz, we isolate the atom by drawing an imaginary sphere of radius R (large compared to the lattice constant a, but small compared to macroscopic variations of $\mathbf{E}$) centered at the atomic position.

The total local field is decomposed into four distinct contributions:

$$\mathbf{E}_{\text{loc}} = \mathbf{E}_0 + \mathbf{E}_1 + \mathbf{E}_2 + \mathbf{E}_3 \tag{8}$$

1. $\mathbf{E}_0$: Electric field produced by external charges on electrodes.
2. $\mathbf{E}_1$: Depolarization field resulting from the polarization surface charge on the external boundary of the dielectric specimen ($\mathbf{E}_1 = -N_{\text{dep}}\mathbf{P}/\epsilon_0$).
3. $\mathbf{E}_2$: Surface charge field on the boundary of the spherical Lorentz cavity.
4. $\mathbf{E}_3$: Microscopic field produced by the individual discrete dipoles located *inside* the Lorentz sphere.

The macroscopic field already accounts for the external charges and external boundaries:

$$\mathbf{E} = \mathbf{E}_0 + \mathbf{E}_1 \tag{9}$$

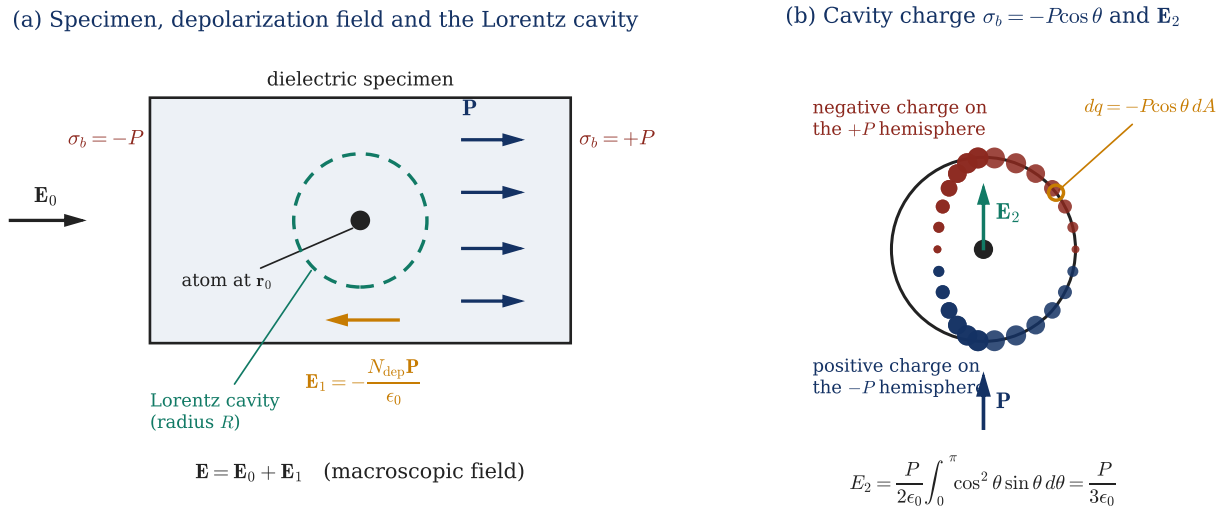


Figure 1: The Lorentz cavity method for computing the microscopic local electric field $\mathbf{E}_{\text{loc}}$.

1.2.2 Evaluation of the Cavity Field $\mathbf{E}_2$

The bound surface charge on the cavity wall is $dq = -P\cos\theta\, dA = -P\cos\theta(2\pi R^2 \sin\theta\, d\theta)$; the minus sign appears because the outward normal of the dielectric points *into* the cavity, so the wall

charge is negative on the $+P$ hemisphere. The field of such an element at the centre is directed along $-\hat{r}$ (towards the element, since $dq < 0$ there), and its component along $\mathbf{P}$ is:

$$dE_2 = -\frac{1}{4\pi\epsilon_0}\frac{dq\cos\theta}{R^2} = \frac{P}{2\epsilon_0}\cos^2\theta\sin\theta\,d\theta \tag{10}$$

Integrating over θ from 0 to π:

$$E_2 = \frac{P}{2\epsilon_0}\int_0^\pi \cos^2\theta\sin\theta\,d\theta = \frac{P}{2\epsilon_0}\left[-\frac{\cos^3\theta}{3}\right]_0^\pi = \frac{\mathbf{P}}{3\epsilon_0} \tag{11}$$

The cavity field therefore *reinforces* the polarization, and is directed along $\mathbf{P}$.

1.2.3 Internal Dipole Field $\mathbf{E}_3$

The field produced at the centre by a point dipole $\mathbf{p}_j$ sitting at $\mathbf{r}_j$ inside the cavity is

$$\mathbf{E}_3 = \frac{1}{4\pi\epsilon_0}\sum_j \frac{3(\mathbf{p}_j\cdot\mathbf{r}_j)\mathbf{r}_j - r_j^2\mathbf{p}_j}{r_j^5} \tag{12}$$

For a lattice site with cubic symmetry (sc, bcc, fcc, or tetrahedral coordination) or isotropic random distributions, the second-rank susceptibility of the site is isotropic, so that

$$\sum_j \frac{x_j^2}{r_j^5} = \sum_j \frac{y_j^2}{r_j^5} = \sum_j \frac{z_j^2}{r_j^5} = \frac{1}{3}\sum_j \frac{r_j^2}{r_j^5}, \qquad \sum_j \frac{x_j y_j}{r_j^5} = 0 \tag{13}$$

The bracketed sum then vanishes identically — the cubic environment is an exact cancellation partner for the dipoles inside the cavity:

$$\mathbf{E}_3 = 0 \tag{14}$$

This cancellation is exact only for cubic or isotropic environments. In a non-cubic crystal $\mathbf{E}_3$ does not vanish, which is precisely why anisotropic permittivity has to be carried as a tensor ϵ_{ij}, and why the scalar Clausius–Mossotti law derived below fails away from cubic symmetry.

Therefore, the **Lorentz Local Field** in a cubic dielectric is:

Lorentz local field

$$\mathbf{E}_{\text{loc}} = \mathbf{E} + \frac{\mathbf{P}}{3\epsilon_0} \quad \text{(SI)} \qquad \Longleftrightarrow \qquad \mathbf{E}_{\text{loc}} = \mathbf{E} + \frac{4\pi}{3}\mathbf{P} \quad \text{(Gaussian)} \tag{15}$$

1.3 Depolarization Fields and the Dependence on Sample Geometry

The depolarization term $\mathbf{E}_1$ of the Lorentz construction is not a universal constant: it is fixed by the *shape* of the specimen. A uniformly polarised body carries bound surface charge $\sigma_b = \mathbf{P}\cdot\hat{\mathbf{n}}$, and the field that this charge produces in the interior of the body is

$$\mathbf{E}_{\text{dep},i} = -\frac{N_i P_i}{\epsilon_0} \quad \text{(SI)} \qquad \Longleftrightarrow \qquad \mathbf{E}_{\text{dep},i} = -4\pi N_i P_i \quad \text{(Gaussian)} \tag{16}$$

where N_i are the dimensionless depolarization factors of the specimen. They are constrained by the trace rule

$$\sum_i N_i = 1 \quad \text{(SI)} \qquad \Longleftrightarrow \qquad \sum_i N_i = 4\pi \quad \text{(Gaussian)} \tag{17}$$

Geometry	**Field direction**	**SI** N_i	**CGS** N_i
Sphere	any axis	$1/3$	$4\pi/3$
Thin plate / slab	normal to the face	1	4π
Thin plate / slab	parallel to the face	0	0
Long needle / cylinder	transverse to the axis	$1/2$	2π
Long needle / cylinder	along the axis	0	0

Table 1: Depolarization factors for the canonical specimen geometries. The trace $\sum_i N_i$ equals 1 in SI and 4π in Gaussian units.

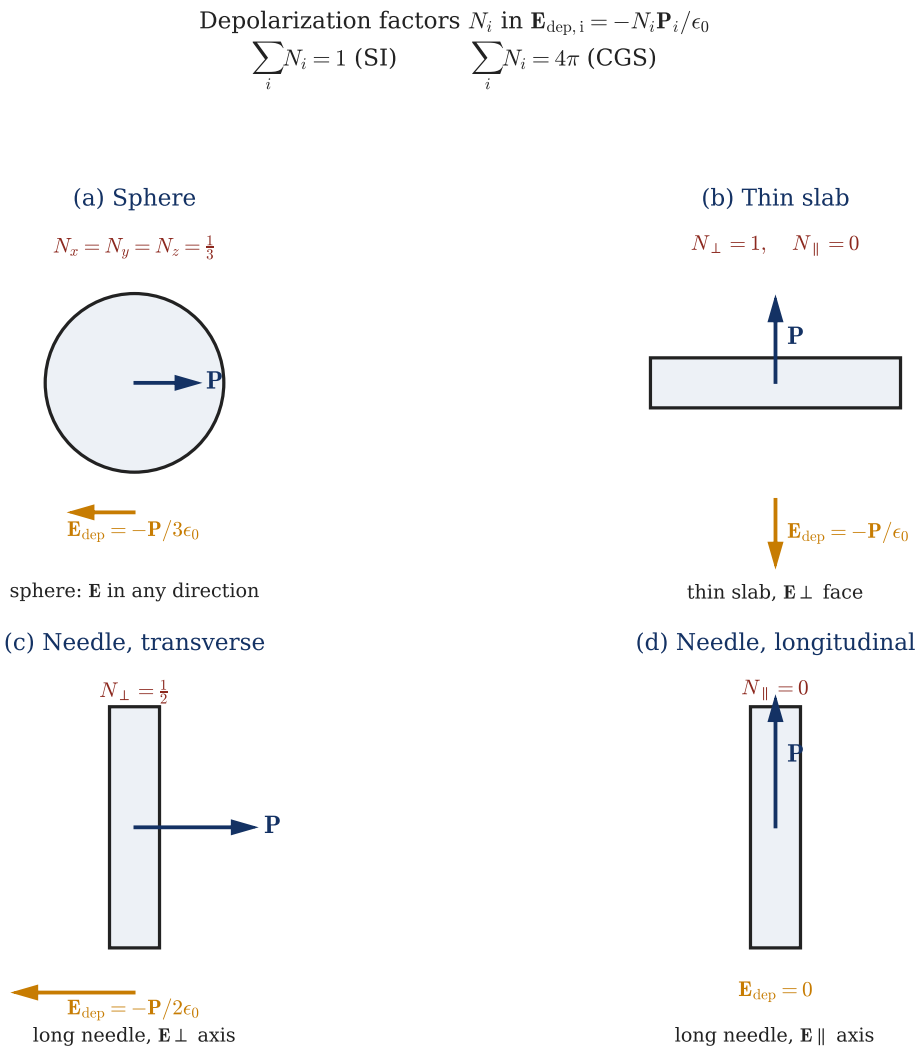


Figure 2: Depolarization fields of the canonical specimen geometries. The depolarizing field always opposes the polarization, and its strength is set by the shape of the sample rather than by the material.

Why the geometry has to be quoted with the measurement: For a thin slab with the field normal to the faces the depolarizing field is as large as the polarization itself ($N = 1$), whereas a long needle aligned with the field has no depolarizing field at all ($N = 0$). Since the local field entering the Clausius–Mossotti relation is $\mathbf{E}_{\text{loc}} = \mathbf{E} - N_i\mathbf{P}_i/\epsilon_0$, a dielectric constant quoted without its specimen geometry is an incomplete specification. This is the reason a slab measured between electrodes and a free crystal of the same material need not return the same ϵ_r.

1.4 Microscopic Polarizability and Clausius–Mossotti

At the microscopic level, an atom experiences $\mathbf{E}_{\text{loc}}$, which induces an atomic dipole moment:

$$\mathbf{p} = \alpha\mathbf{E}_{\text{loc}} \tag{18}$$

where α is the atomic polarizability. For a substance with concentration N atoms per unit volume:

$$\mathbf{P} = N\mathbf{p} = N\alpha\mathbf{E}_{\text{loc}} = N\alpha\left(\mathbf{E} + \frac{\mathbf{P}}{3\epsilon_0}\right) \tag{19}$$

Rearranging terms for $\mathbf{P}$:

$$\mathbf{P}\left(1 - \frac{N\alpha}{3\epsilon_0}\right) = N\alpha\mathbf{E} \implies \mathbf{P} = \frac{N\alpha}{1 - \frac{N\alpha}{3\epsilon_0}}\mathbf{E} \tag{20}$$

Equating this with $\mathbf{P} = \epsilon_0(\epsilon_r - 1)\mathbf{E}$, we obtain:

$$\epsilon_0(\epsilon_r - 1) = \frac{N\alpha}{1 - \frac{N\alpha}{3\epsilon_0}} \tag{21}$$

Dividing both sides by $\epsilon_0(\epsilon_r + 2)$ and simplifying yields the celebrated **Clausius-Mossotti relation**:

$$\frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{N\alpha}{3\epsilon_0} = \frac{1}{3\epsilon_0}\sum_j N_j\alpha_j \tag{22}$$

Expressing the dielectric constant explicitly:

$$\epsilon_r = \frac{1 + \frac{2N\alpha}{3\epsilon_0}}{1 - \frac{N\alpha}{3\epsilon_0}} \tag{23}$$

Physical Consequence — Polarization Catastrophe: Notice that if $N\alpha/(3\epsilon_0) \to 1$, the denominator approaches zero, leading to $\epsilon_r \to \infty$. This implies that a finite spontaneous polarization $\mathbf{P}$ can exist even in the absence of an applied macroscopic field $\mathbf{E} = 0$. This instability is the classic electrostatic criterion for the onset of **ferroelectricity**.

Sanity check — the gas limit is recovered: The Clausius–Mossotti relation must collapse to the dilute-gas baseline of Subsection 1.1 when the local field is negligible. Writing $x \equiv N\alpha/(3\epsilon_0)$, the exact result is

$$\epsilon_r - 1 = \frac{3x}{1 - x} = 3x + O(x^2) \tag{24}$$

which is precisely the gas law $\epsilon_r = 1 + N\alpha/\epsilon_0$ to first order. The entire content of the local-field construction is therefore the higher-order terms.

Unit conventions (SI and Gaussian)

$$\mathbf{D} = \epsilon_0\mathbf{E} + \mathbf{P}, \quad \epsilon_r = 1 + \chi_e \quad \text{(SI)} \quad \Longleftrightarrow \quad \mathbf{D} = \mathbf{E} + 4\pi\mathbf{P} \quad \text{(Gaussian)} \quad (25)$$

$$\alpha_e = 4\pi\epsilon_0 R^3 \quad \text{(SI)} \quad \Longleftrightarrow \quad \alpha_e = R^3 \quad \text{(Gaussian)} \quad (26)$$

$$\mathbf{E}_{\text{loc}} = \mathbf{E} + \frac{\mathbf{P}}{3\epsilon_0} \quad \text{(SI)} \quad \Longleftrightarrow \quad \mathbf{E}_{\text{loc}} = \mathbf{E} + \frac{4\pi}{3}\mathbf{P} \quad \text{(Gaussian)} \quad (27)$$

$$\frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{N\alpha}{3\epsilon_0} \quad \text{(SI)} \quad \Longleftrightarrow \quad \frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{4\pi}{3}N\alpha \quad \text{(Gaussian)} \quad (28)$$

1.5 The Lorentz–Lorenz Relation and a Static-versus-Optical Diagnostic

At optical frequencies the ions are far too massive to follow the oscillating field, so only the electronic polarizability survives and the measured response is the optical one. For a non-magnetic medium Maxwell's relation gives

$$\epsilon_r(\omega_{\text{opt}}) = n^2 \quad (29)$$

where n is the refractive index. Substituting this into the Clausius–Mossotti relation with $\alpha \to \alpha_e$ produces the **Lorentz–Lorenz equation**:

$$\frac{n^2 - 1}{n^2 + 2} = \frac{N\alpha_e}{3\epsilon_0} \quad (30)$$

This is a genuinely useful experimental diagnostic, because it separates the two contributions that a static measurement mixes together: the optical measurement sees only α_e, the static measurement sees $\alpha_e + \alpha_i + \alpha_d$.

Case study: sodium chloride against diamond.

- **Sodium Chloride (NaCl)** — an ionic crystal:
  - static (low-frequency) relative permittivity $\epsilon_r(0) \approx 5.62$,
  - optical refractive index $n \approx 1.50$, so $n^2 \approx 2.25$,
  - since $\epsilon_r(0) \gg n^2$, a large share of the static response comes from the ionic polarizability α_i associated with the relative displacement of Na^+ and Cl^- sublattices, which simply cannot follow an optical field.
- **Diamond (C)** — a covalent crystal:
  - static $\epsilon_r(0) \approx 5.68$,
  - optical $n \approx 2.38$, so $n^2 \approx 5.66$,
  - here $\epsilon_r(0) \approx n^2$: the bonding carries no permanent ionic charge, so the polarization is exclusively electronic in both frequency regimes.

Quantitatively, insert both numbers into the Clausius–Mossotti factor $Y \equiv (\epsilon_r - 1)/(\epsilon_r + 2)$:

$$Y^{(n)}_{\text{NaCl}} = \frac{1.50^2 - 1}{1.50^2 + 2} = 0.294 \qquad \text{but} \qquad Y^{(\text{static})}_{\text{NaCl}} = \frac{5.62 - 1}{5.62 + 2} = 0.606 \quad (31)$$

$$Y^{(n)}_{\text{diamond}} = \frac{2.38^2 - 1}{2.38^2 + 2} = 0.609 \qquad \text{and} \qquad Y^{(\text{static})}_{\text{diamond}} = \frac{5.68 - 1}{5.68 + 2} = 0.609 \quad (32)$$

The two numbers agree to three digits for diamond (purely electronic) and differ by a factor of two for NaCl. Equivalently, the ionic share of the static response of NaCl is

$$1 - \frac{n^2}{\epsilon_r(0)} = 1 - \frac{2.25}{5.62} = 0.60, \tag{33}$$

i.e. about 60% of the static polarization of rock salt is ionic, against essentially 0% for diamond (less than 1%).

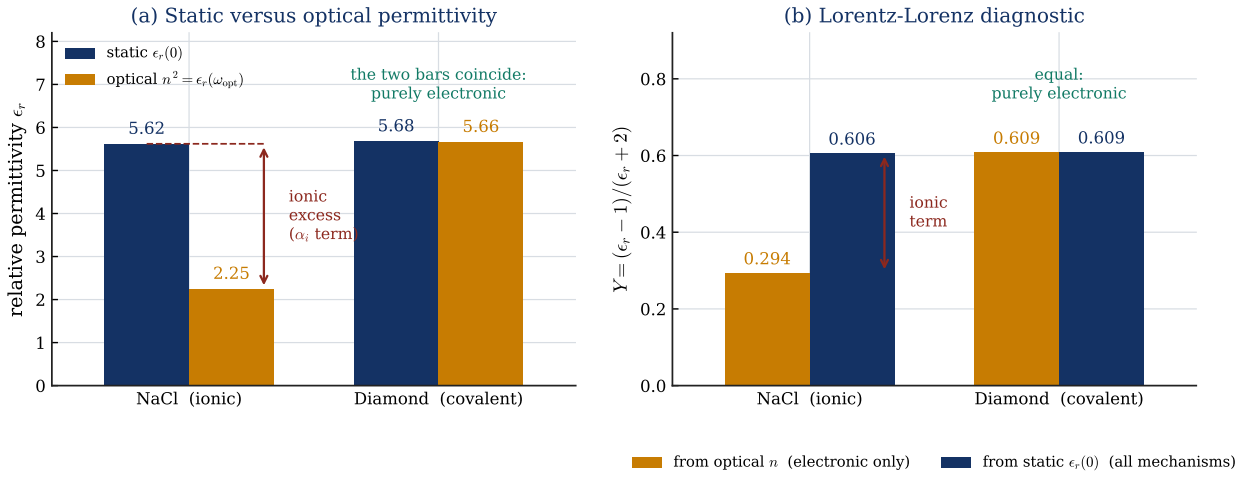


Figure 3: The static-versus-optical diagnostic. NaCl shows a large ionic excess between $\epsilon_r(0)$ and n^2; for diamond the two coincide, so the response is purely electronic.

1.6 Microscopic Mechanisms of Polarization and Frequency Response

The total atomic polarizability α is a sum of three distinct physical contributions:

$$\alpha_{\text{total}} = \alpha_e + \alpha_i + \alpha_d \tag{34}$$

1. **Electronic Polarizability (α_e):** Displacement of the valence electron cloud relative to the positive nucleus.

   - **Thomson atom.** Model an atom of atomic number Z as a nucleus $+Ze$ embedded in a spherically symmetric, rigid electron cloud of charge $-Ze$ and radius R with uniform density

   $$\rho = \frac{-Ze}{\frac{4}{3}\pi R^3} \tag{35}$$

   In a local field E_{loc} the nucleus is displaced by $+x$ relative to the geometric centre of the cloud. Constructing a spherical Gaussian surface of radius x concentric with the displaced nucleus, the enclosed charge is

   $$q_{\text{enc}} = (-Ze)\frac{x^3}{R^3} \tag{36}$$

   and Gauss's law gives $E_r(x) \cdot 4\pi x^2 = q_{\text{enc}}/\epsilon_0$. The resulting Coulomb force pulls the displaced nucleus back towards the centre of the cloud, with magnitude

   $$F_{\text{res}} = (Ze)\,E_r(x) = \frac{(Ze)^2 x}{4\pi\epsilon_0 R^3} \tag{37}$$

Balancing this against the driving force $(Ze)E_{\text{loc}}$ fixes the equilibrium displacement

$$x = \frac{4\pi\epsilon_0 R^3}{Ze} E_{\text{loc}} \tag{38}$$

and hence the induced dipole moment $p = (Ze)x$ and the electronic polarizability

$$\alpha_e = \frac{p}{E_{\text{loc}}} = 4\pi\epsilon_0 R^3 \quad \text{(SI)} \qquad \Longleftrightarrow \qquad \alpha_e = R^3 \quad \text{(Gaussian)} \tag{39}$$

The electronic polarizability is therefore just the atomic volume up to the factor $4\pi\epsilon_0$: larger atoms polarise more.

- If instead a single effective electron is displaced inside a Z-electron cloud, the same algebra with charge $-e$ gives $\alpha_e = 4\pi\epsilon_0 R_0^3/Z$. The electronic response is temperature-independent and survives up to ultraviolet frequencies ($\omega \sim 10^{15}$–10^{16} rad s^{-1}).

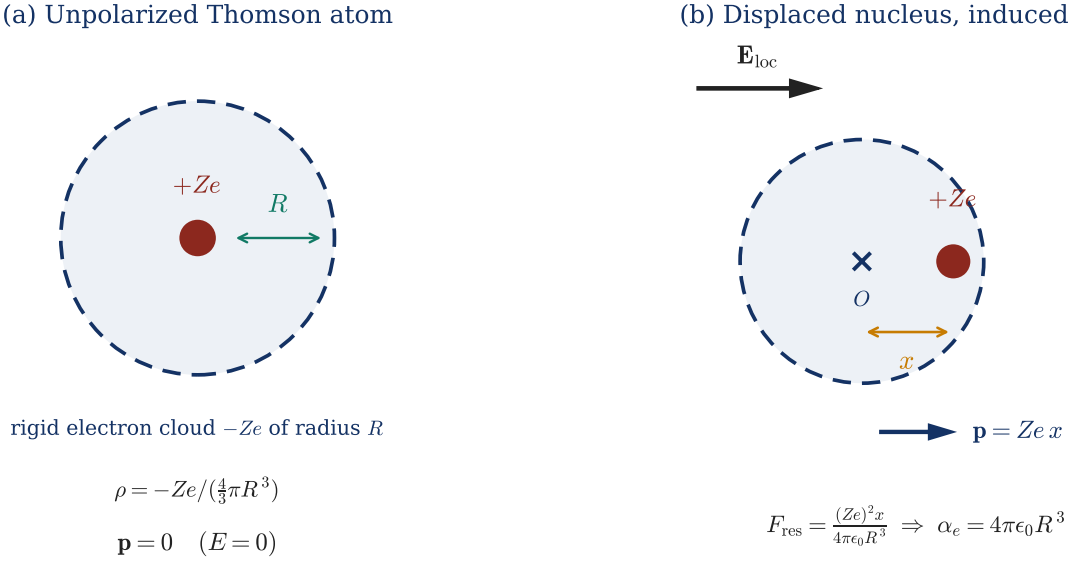


Figure 4: The Thomson atom. Displacing the nucleus against the restoring force of its own electron cloud is the origin of the electronic polarizability $\alpha_e = 4\pi\epsilon_0 R^3$.

2. **Ionic (Atomic) Polarizability (α_i):** Relative displacement of positive and negative sublattices in ionic crystals (e.g., Na^+ and Cl^-).

   - For ions with reduced mass μ and optical vibration frequency ω_0:

   $$\alpha_i = \frac{q^2}{\mu\omega_0^2} \tag{40}$$

   - Responds in the infrared regime ($\omega \sim 10^{12}$–10^{13} rad s^{-1}).

3. **Dipolar (Orientational) Polarizability (α_d):** Thermal reorientation of permanent electric dipoles $\mathbf{p}_0$ (e.g., H_2O, polar molecules).

   - In an electric field $\mathbf{E}$, the potential energy is $U = -\mathbf{p}_0 \cdot \mathbf{E} = -p_0 E \cos\theta$.
   - By Boltzmann statistics, the average dipole moment along $\mathbf{E}$ is $\langle p_z \rangle = p_0 L(x)$ where $L(x) = \coth x - \frac{1}{x}$ is the Langevin function ($x = p_0 E/k_B T$).
   - In the weak-field limit ($p_0 E \ll k_B T$):

   $$\langle p_z \rangle \approx \frac{p_0^2}{3k_B T} E_{\text{loc}} \implies \alpha_d = \frac{p_0^2}{3k_B T} \tag{41}$$

- Strongly temperature-dependent ($\propto 1/T$) and sluggish; it freezes out at radio and microwave frequencies ($\omega \lesssim 10^9$ rad s^{-1}).

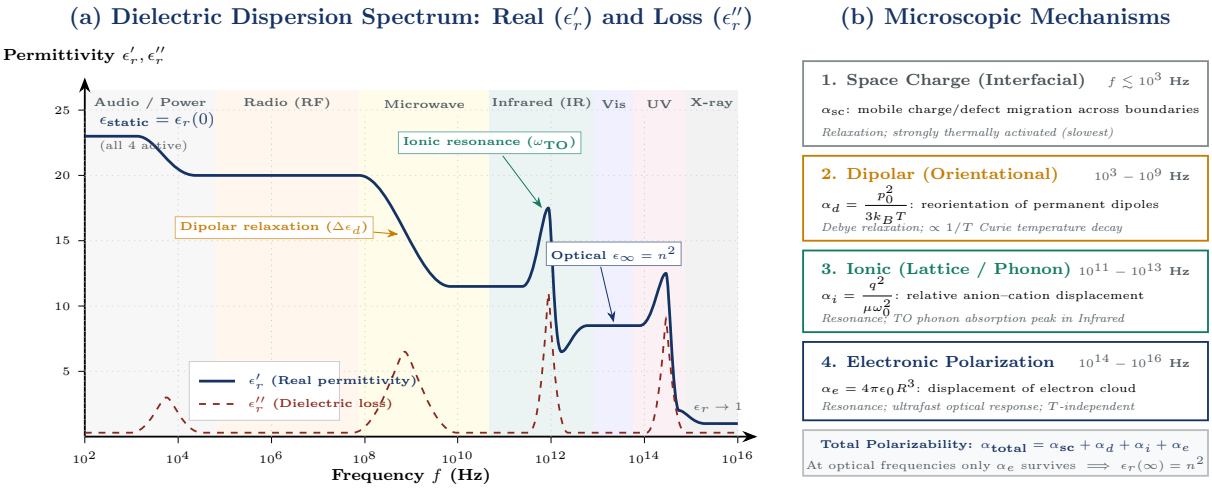


Figure 5: Frequency dispersion of the dielectric polarization mechanisms.

From static polarizability to a frequency-dependent response. The three mechanisms are separated in a frequency sweep because each one is frozen out once the driving frequency exceeds its own relaxation rate: the dipolar term dies out at radio frequencies, the ionic term in the infrared, and only the electronic term survives into the visible. The dynamic counterpart of the Thomson atom — an electron on a damped harmonic oscillator, from which the complex dielectric function follows — is derived in Subsection 2.4 of Lecture 2.

1.7 Piezoelectric Materials

1.7.1 Phenomenology and Symmetry Criteria

Definition: Piezoelectricity is the linear coupling between mechanical stress and electric polarization in non-centrosymmetric materials.

- **Direct Piezoelectric Effect:** Applying a mechanical stress σ_{jk} induces a dielectric polarization P_i:
$$P_i = d_{ijk}\sigma_{jk} \tag{42}$$
where d_{ijk} is the third-rank piezoelectric strain coefficient tensor.
- **Converse Piezoelectric Effect:** Applying an electric field E_k induces a mechanical strain ϵ_{ij}:
$$\epsilon_{ij} = d_{kij}E_k \tag{43}$$

Centrosymmetry Elimination Rule: Under spatial inversion $\mathbf{r} \to -\mathbf{r}$:

- The polarization vector is odd: $P_i \to -P_i$.
- The stress tensor is even: $\sigma_{jk} \to \sigma_{jk}$.

Thus: $-P_i = d_{ijk}\sigma_{jk} \implies d_{ijk} = -d_{ijk} \implies d_{ijk} \equiv 0$. **Conclusion:** A crystal possessing a center of inversion cannot be piezoelectric. Of the 32 crystal point groups, 21 lack inversion symmetry, and 20 of these exhibit piezoelectricity (group 432 is the only non-centrosymmetric exception due to high octahedral symmetry).

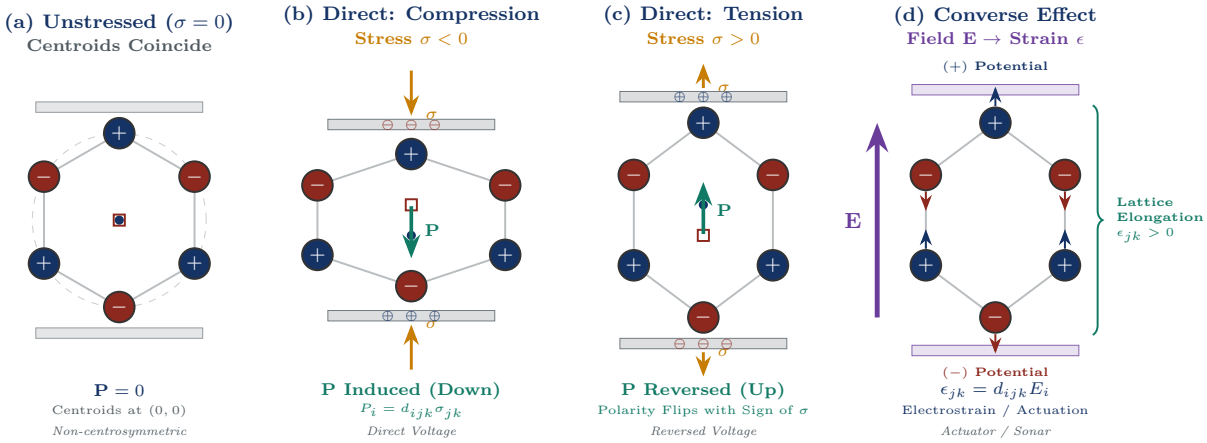


Figure 6: Microscopic origin of the direct and converse piezoelectric effects.

1.7.2 Key Materials and Applications

- **α-Quartz (SiO_2):** Non-ferroelectric piezoelectric; highly stable mechanical Q-factor ($> 10^5$), utilized in crystal oscillators for clocks, microprocessors, and frequency filters.
- **Lead Zirconate Titanate (PZT, $PbZr_{1-x}Ti_xO_3$):** A perovskite solid solution showing an enormous piezoelectric response near the morphotropic phase boundary ($x \approx 0.48$). Extensively used in ultrasound transducers, SONAR, and micro-positioning actuators.
- **Polyvinylidene Fluoride (PVDF):** Semi-crystalline piezo-polymer used in flexible wearable sensors and hydrophones.

2 Ferroelectric Materials and Optical Properties of Solids

Duration: 90 Minutes — Walkthrough Target: Spontaneous Order, Phase Transitions, Perovskites, Complex Permittivity, and Optical Constants.

2.1 Ferroelectricity: Fundamentals and Classification

Definition: A ferroelectric material possesses a spontaneous dielectric polarization $\mathbf{P}_s$ in the absence of an applied electric field, and this polarization can be reversed or reoriented between two or more stable states by an external electric field.

Hierarchy of Dielectric Classes:

$$\begin{aligned} \text{All dielectrics} &\supset \text{piezoelectrics (20 classes)} \\ &\supset \text{pyroelectrics (10 polar classes)} \\ &\supset \text{ferroelectrics (switchable } \mathbf{P}_s) \end{aligned} \tag{44}$$

2.1.1 Curie-Weiss Law

Above the ferroelectric Curie temperature T_c, the material transitions into a non-polar paraelectric state. In this paraelectric regime ($T > T_c$), the static dielectric constant diverges according to the **Curie-Weiss Law**:

$$\epsilon_r = \epsilon_\infty + \frac{C}{T - T_0} \approx \frac{C}{T - T_0} \tag{45}$$

where C is the Curie constant (10^3–10^5 K) and T_0 is the Curie-Weiss temperature ($T_0 \leq T_c$).

2.1.2 Classification by Microscopic Transition Mechanism

1. **Displacive Ferroelectrics (e.g., Perovskites: $BaTiO_3$, $PbTiO_3$):**
   - In the paraelectric phase ($T > 120°C$ for $BaTiO_3$), the crystal is cubic ($Pm\bar{3}m$): Ba^{2+} at cube corners, O^{2-} at face centers, Ti^{4+} at body center.
   - Below T_c, Ti^{4+} and Ba^{2+} shift relative to the oxygen octahedral cage, breaking cubic symmetry to tetragonal ($P4mm$), generating a spontaneous dipole moment ($P_s \approx 26\,\mu C/cm^2$).
2. **Order-Disorder Ferroelectrics (e.g., Potassium Dihydrogen Phosphate, KH_2PO_4 / KDP):**
   - Protons sit in double-well potential wells along $O - H \cdots O$ hydrogen bonds.
   - For $T > T_c$, protons hop dynamically between wells (disordered state). For $T < T_c$, protons freeze cooperatively into one of the off-center minima.
   - **Deuteration Effect:** Substituting H with D (KD_2PO_4) increases T_c from 123 K to 213 K (a shift of 90 K with $< 2\%$ mass change), evidencing quantum tunneling of protons through the double-well barrier.

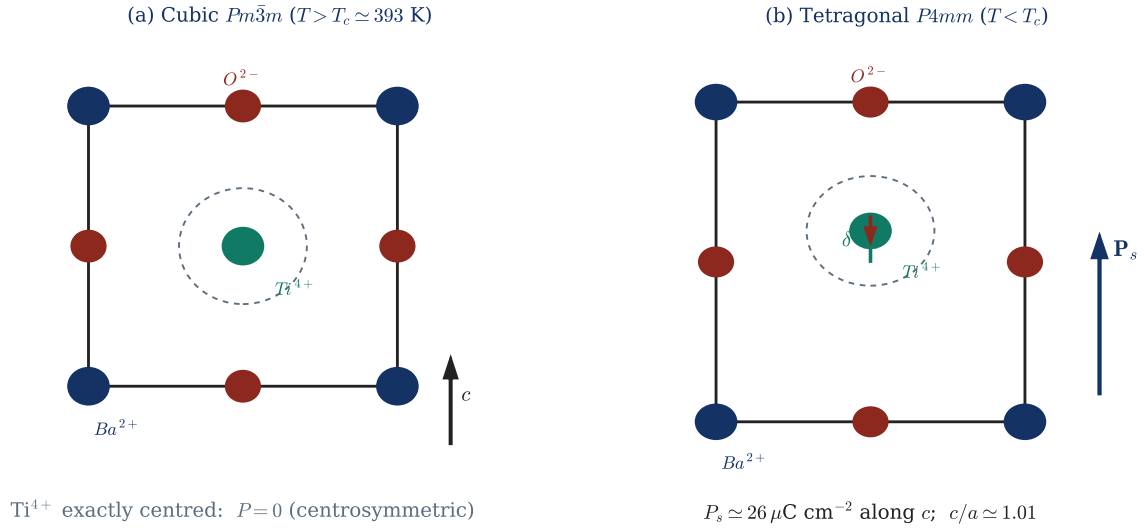


Figure 7: The cubic-to-tetragonal phase transition in barium titanate ($BaTiO_3$).

2.2 Ferroelectric Domains and Hysteresis

To minimize depolarizing electrostatic field energy and elastic strain energy, ferroelectrics break into regions of uniform polarization called **ferroelectric domains**, separated by domain walls (typically 180° and 90° walls in perovskites).

2.2.1 The P–E Hysteresis Loop

Application of a cyclic AC electric field results in characteristic hysteresis due to domain wall pinning, nucleation, and domain reorientation:

- **Spontaneous Polarization (P_s):** Extrapolation of the saturated high-field polarization curve back to $E = 0$.
- **Remanent Polarization (P_r):** Net polarization remaining in the sample when the external field is removed ($E = 0$).
- **Coercive Field (E_c):** The reverse field required to reduce the net polarization to zero ($P = 0$).

2.2.2 Lattice Dynamics: Cochran Soft Phonon Mode Theory

According to W. Cochran, a displacive ferroelectric transition represents a dynamical instability of the transverse optical (TO) phonon branch at zone center ($q = 0$). The Lyddane-Sachs-Teller (LST) relation states:

$$\frac{\omega_{LO}^2}{\omega_{TO}^2} = \frac{\epsilon(0)}{\epsilon_\infty} \tag{46}$$

Combining this with the Curie-Weiss law $\epsilon(0) \propto (T - T_c)^{-1}$ leads directly to:

$$\omega_{TO}^2(T) = \gamma(T - T_c) \implies \omega_{TO} \to 0 \quad \text{as } T \to T_c^+ \tag{47}$$

As temperature cools toward T_c, the restoring force for this optical phonon mode softens to zero; the lattice displacement freezes permanently into the unit cell, resulting in the ferroelectric structure.

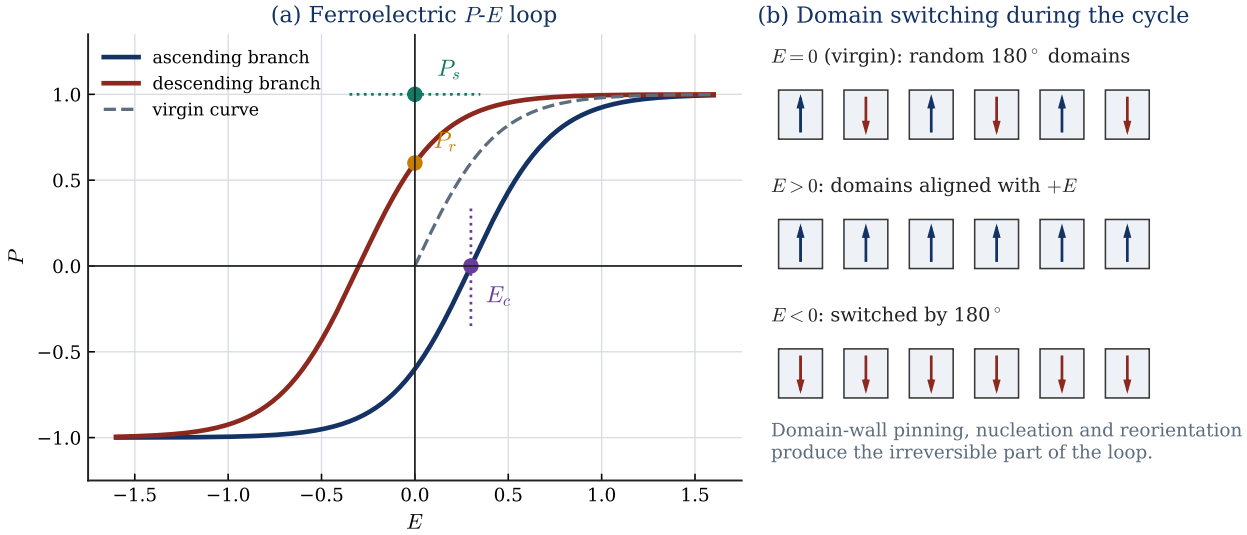


Figure 8: Ferroelectric hysteresis loop and domain switching behavior under external field.

2.3 Optical Constants and Complex Dielectric Function

2.3.1 Maxwell Equations and the Complex Refractive Index

Consider a monochromatic plane electromagnetic wave propagating along the $+z$ direction in an absorbing solid:

$$\mathbf{E}(z,t) = \mathbf{E}_0 e^{i(\tilde{k}z-\omega t)}, \quad \tilde{k} = \frac{\omega}{c}\tilde{n}(\omega) \tag{48}$$

where $\tilde{n}(\omega)$ is the **complex refractive index**:

$$\tilde{n}(\omega) = n(\omega) + i\kappa(\omega) \tag{49}$$

Here, n is the real refractive index (phase velocity $v_p = c/n$) and κ is the extinction coefficient. Substituting $\tilde{n}$ into the wave:

$$\mathbf{E}(z,t) = \mathbf{E}_0 e^{-\frac{\omega\kappa}{c}z} e^{i\left(\frac{\omega n}{c}z-\omega t\right)} \tag{50}$$

The optical intensity $I(z) \propto |\mathbf{E}|^2$ decays as:

$$I(z) = I_0 e^{-\alpha z}, \quad \text{where } \alpha(\omega) = \frac{2\omega\kappa(\omega)}{c} = \frac{4\pi\kappa(\omega)}{\lambda_0} \tag{51}$$

$\alpha(\omega)$ is the **optical absorption coefficient** (Beer-Lambert law).

2.3.2 Relation between $\tilde{n}$ and the Complex Dielectric Function $\tilde{\epsilon}$

From Maxwell's wave equation, $\tilde{k}^2 = \frac{\omega^2}{c^2}\tilde{\epsilon}_r(\omega)$, hence:

$$\tilde{\epsilon}_r(\omega) = \epsilon_1(\omega) + i\epsilon_2(\omega) = \tilde{n}^2 = (n+i\kappa)^2 = (n^2-\kappa^2) + i(2n\kappa) \tag{52}$$

Equating real and imaginary parts:

$$\epsilon_1(\omega) = n^2 - \kappa^2 \tag{53}$$

$$\epsilon_2(\omega) = 2n\kappa \tag{54}$$

Inverting these relations gives the experimental optical constants in terms of the dielectric function:

$$n(\omega) = \sqrt{\frac{\sqrt{\epsilon_1^2 + \epsilon_2^2} + \epsilon_1}{2}} \tag{55}$$

$$\kappa(\omega) = \sqrt{\frac{\sqrt{\epsilon_1^2 + \epsilon_2^2} - \epsilon_1}{2}} \tag{56}$$

2.3.3 Normal Incidence Reflectance

From electromagnetic boundary conditions at a planar vacuum-dielectric interface:

$$\tilde{r}(\omega) = \frac{\tilde{n} - 1}{\tilde{n} + 1} = \frac{n + i\kappa - 1}{n + i\kappa + 1} \tag{57}$$

The reflectance $R(\omega) = |\tilde{r}(\omega)|^2$ is given by the Fresnel formula:

$$R(\omega) = \frac{(n-1)^2 + \kappa^2}{(n+1)^2 + \kappa^2} \tag{58}$$

When the material is strongly absorbing or when $\epsilon_1 < 0$ (such that $\kappa \gg n$), the reflectance $R \to 1$ (100% reflection, characteristic of metallic and Reststrahlen bands).

2.4 Microscopic Theory: The Lorentz Oscillator Model

Model an electron bound to an atomic core with natural resonance frequency ω_0 and phenomenological damping rate γ:

$$m_e \left(\frac{d^2x}{dt^2} + \gamma \frac{dx}{dt} + \omega_0^2 x \right) = -eE_0 e^{-i\omega t} \tag{59}$$

Solving for the steady-state displacement $x(t)$:

$$x(t) = -\frac{e/m_e}{\omega_0^2 - \omega^2 - i\gamma\omega} E(t) \tag{60}$$

The induced dipole moment is $p = -ex(t)$, yielding the polarization $P = Np = \epsilon_0(\tilde{\epsilon}_r - 1)E$:

$$\tilde{\epsilon}_r(\omega) = 1 + \frac{Ne^2}{\epsilon_0 m_e} \frac{1}{\omega_0^2 - \omega^2 - i\gamma\omega} = 1 + \frac{\omega_p^2}{\omega_0^2 - \omega^2 - i\gamma\omega} \tag{61}$$

where $\omega_p = \sqrt{\frac{Ne^2}{\epsilon_0 m_e}}$ is the plasma frequency.

Separating into real and imaginary parts:

$$\epsilon_1(\omega) = 1 + \omega_p^2 \frac{\omega_0^2 - \omega^2}{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2} \tag{62}$$

$$\epsilon_2(\omega) = \omega_p^2 \frac{\gamma\omega}{(\omega_0^2 - \omega^2)^2 + \gamma^2\omega^2} \tag{63}$$

Physical Regimes:

- **Normal Dispersion ($\frac{d\epsilon_1}{d\omega} > 0$):** Far from resonance ($\omega \ll \omega_0$ or $\omega \gg \omega_0$), absorption ϵ_2 is negligible.
- **Anomalous Dispersion ($\frac{d\epsilon_1}{d\omega} < 0$):** Inside the resonance window $|\omega - \omega_0| \lesssim \gamma$, ϵ_1 decreases with frequency, accompanied by a sharp absorption peak in $\epsilon_2(\omega)$.

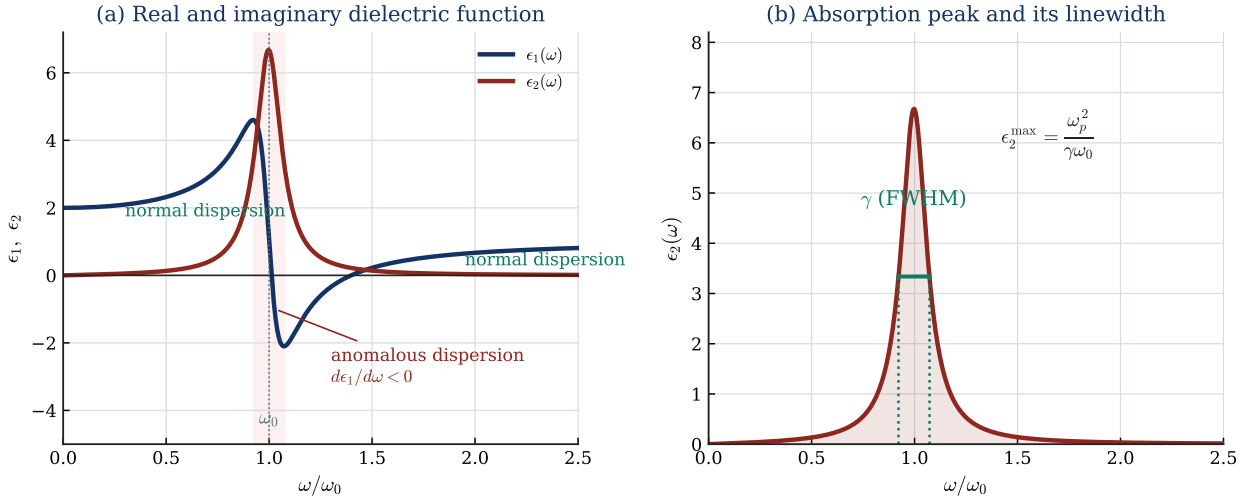


Figure 9: Real and imaginary parts of the dielectric function in the Lorentz oscillator model.

2.4.1 Numerical Recipe

The two curves of Figure 9 follow directly from the two expressions above. The script below is the minimal version of `make_figures.py` that ships the figure, and is worth putting on the screen once so that students see that ϵ_1 and ϵ_2 are not independent curves but the two projections of a single complex function:

Listing 1: Lorentz oscillator: real and imaginary parts of the dielectric function.

```
import numpy as np
import matplotlib.pyplot as plt

omega_0 = 10.0        # natural resonance frequency (rad/s)
gamma   = 1.0         # phenomenological damping rate
wp2     = 40.0        # plasma frequency squared: N e^2 / (eps0 m)

omega = np.linspace(0, 20, 1000)
denom = (omega_0**2 - omega**2)**2 + (gamma*omega)**2

eps_1 = 1.0 + wp2*(omega_0**2 - omega**2)/denom    # real part
eps_2 = wp2*(gamma*omega)/denom                    # imaginary part

plt.plot(omega, eps_1, 'b-', lw=2, label=r'$\varepsilon_1(\omega)$')
plt.plot(omega, eps_2, 'r--', lw=2, label=r'$\varepsilon_2(\omega)$')
plt.axvline(omega_0, color='gray', linestyle=':', label=r'$\omega_0$')
plt.xlabel(r'Frequency $\omega$ [rad/s]')
plt.ylabel(r'Permittivity $\varepsilon(\omega)$')
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig('dielectric_dispersion.pdf')
```

2.5 Interband Transitions, Excitons, and Reststrahlen

2.5.1 Direct and Indirect Interband Transitions

In crystalline semiconductors and insulators, optical absorption occurs when photon energy $\hbar\omega \geq E_g$:

- **Direct Absorption:** The conduction band minimum and valence band maximum lie at the same point in the Brillouin zone ($k_c = k_v$, usually Γ at $k = 0$). Because photon momentum $q_{\text{photon}} \approx 0$, vertical transitions occur without phonon assistance:

$$\alpha_{\text{direct}}(\hbar\omega) \propto (\hbar\omega - E_g)^{1/2} \tag{64}$$

  (e.g., GaAs, InSb; sharp absorption edge).

- **Indirect Absorption:** Band extrema are located at different wavevectors ($k_c \neq k_v$). A transition requires emission or absorption of a lattice phonon ($\hbar\Omega_q, \mathbf{q} = \mathbf{k}_c - \mathbf{k}_v$) to conserve crystal momentum:

$$\hbar\omega = E_g + \Delta E_{\text{electron-hole}} \pm \hbar\Omega_q \tag{65}$$

$$\alpha_{\text{indirect}}(\hbar\omega) \propto \frac{(\hbar\omega - E_g \mp \hbar\Omega_q)^2}{e^{\pm\hbar\Omega_q/k_BT} - 1} \tag{66}$$

  (e.g., Si, Ge; gradual absorption threshold).

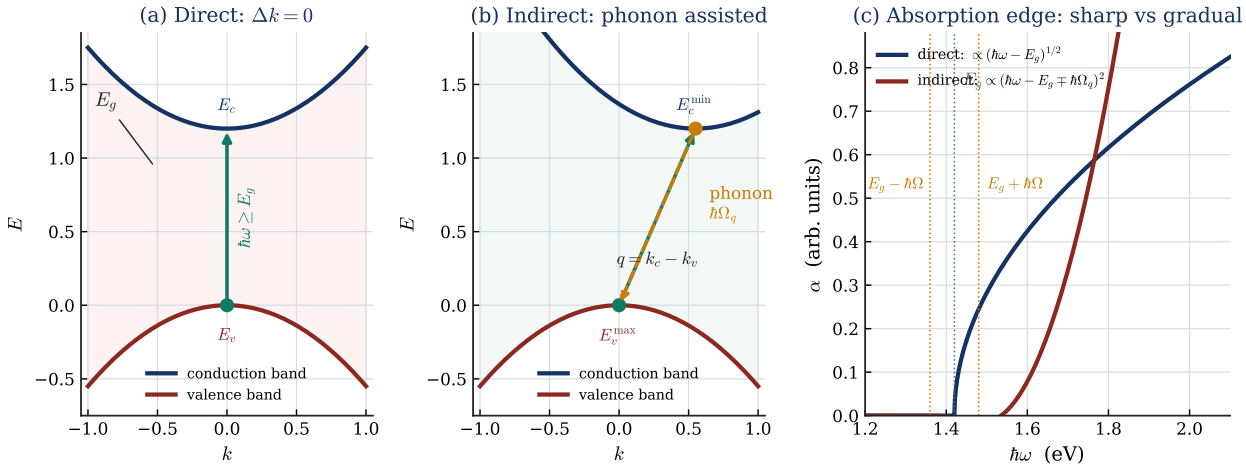


Figure 10: Direct versus indirect optical absorption mechanisms in semiconductors.

2.5.2 Excitons

An exciton is a neutral bound state of an excited electron and a hole interacting via the Coulomb potential screened by the dielectric constant ϵ_r.

- **Wannier-Mott Excitons:** Weakly bound ($E_b \sim$ 5–30 meV), large Bohr radius ($a_{\text{exc}} \gg a_{\text{lattice}}$), common in semiconductors (Si, GaAs). The energy spectrum resembles a hydrogenic atom:

$$E_n = E_g - \frac{\mu^* e^4}{32\pi^2\epsilon_0^2\epsilon_r^2\hbar^2}\frac{1}{n^2} = E_g - \frac{R_y^*}{n^2}, \quad n = 1, 2, 3, \ldots \tag{67}$$

  where $\mu^* = (m_e^{*-1} + m_h^{*-1})^{-1}$ is the reduced effective mass. Sharp excitonic absorption lines appear immediately below the band edge E_g.

- **Frenkel Excitons:** Strongly bound ($E_b \sim$ 0.1–1 eV), tightly localized within a single unit cell, characteristic of molecular crystals and alkali halides.

2.5.3 Reststrahlen Band (Residual Rays)

In ionic crystals (e.g., NaCl), optical phonons couple to transverse electromagnetic waves, creating mixed mode quasiparticles known as **phonon polaritons**. In the frequency window between ω_{TO} and ω_{LO}:

$$\epsilon_1(\omega) = \epsilon_\infty + \frac{\epsilon(0) - \epsilon_\infty}{1 - (\omega/\omega_{TO})^2} < 0 \tag{68}$$

When $\epsilon_1 < 0$ and damping is low, the refractive index is purely imaginary ($n \to 0, \kappa > 0$), resulting in nearly total reflection ($R \approx 100\%$). This frequency region ($\omega_{TO} < \omega < \omega_{LO}$) is the **Reststrahlen band**.

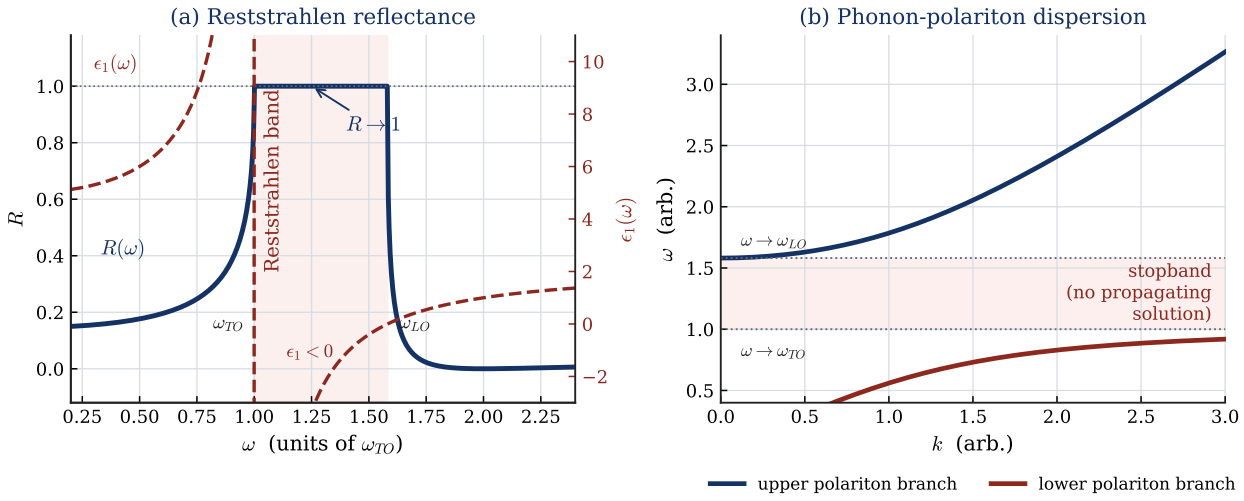


Figure 11: Reststrahlen reflectance band and phonon-polariton dispersion in ionic crystals.

3 Master Summary of Governing Equations and Conceptual Framework

3.1 Consolidated Mathematical Backbone

Table 2 synthesizes the governing formulas across the dielectric, ferroelectric, and optical domains. Every equation is formulated in standard SI units.

Phenomenon	Core Equation	Key Physical Quantities
Dilute-Gas Limit	$\epsilon_r = 1 + \frac{N\alpha}{\epsilon_0}$	N: number density, α: atomic polarizability
Lorentz Local Field	$\mathbf{E}_{\text{loc}} = \mathbf{E} + \frac{\mathbf{P}}{3\epsilon_0}$	$\mathbf{E}$: macroscopic field, $\mathbf{P}$: polarization
Depolarization Field	$\mathbf{E}_{\text{dep},i} = -N_i \mathbf{P}_i/\epsilon_0,\ \sum_i N_i = 1$	N_i: depolarization factor (specimen geometry)
Clausius-Mossotti	$\frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{1}{3\epsilon_0}\sum N_i \alpha_i$	N: atomic density, α: microscopic polarizability
Thomson Atom	$\alpha_e = 4\pi\epsilon_0 R^3$ (SI)	R: atomic radius; rigid electron cloud of $-Ze$
Lorentz-Lorenz	$\frac{n^2-1}{n^2+2} = \frac{N\alpha_e}{3\epsilon_0}$	n: optical refractive index; electronic term only
Piezoelectricity	$P_i = d_{ijk}\sigma_{jk}, \quad \epsilon_{ij} = d_{kij}E_k$	d: piezoelectric tensor, σ: stress, ϵ: strain
Curie-Weiss Law	$\epsilon_r \approx \frac{C}{T - T_0}$ $(T > T_c)$	C: Curie constant, T_0: Curie-Weiss temperature
Lyddane-Sachs-Teller	$\frac{\omega_{LO}^2}{\omega_{TO}^2} = \frac{\epsilon(0)}{\epsilon_\infty}$	ω_{LO}, ω_{TO}: optical phonon frequencies
Optical Complex Index	$\tilde{n} = n + i\kappa = \sqrt{\epsilon_1 + i\epsilon_2}$	n: refraction index, κ: extinction coefficient
Fresnel Normal Reflectance	$R = \frac{(n-1)^2+\kappa^2}{(n+1)^2+\kappa^2}$	R: fraction of reflected power
Exciton Energy Levels	$E_n = E_g - \frac{R_y^*}{n^2}$	R_y^*: effective Rydberg energy, n: quantum number

Table 2: Comprehensive recap of the mathematical backbone for Lectures 1 and 2.

3.2 Conceptual Synthesis and Physical Interconnections

Understanding Module 5B requires appreciating how microscopic interactions dictate macroscopic physical observables:

- **The Feedback Mechanism of the Local Field:** In dense media, induced dipoles enhance the field acting on their neighbors ($\mathbf{E}_{\text{loc}} = \mathbf{E} + \mathbf{P}/3\epsilon_0$). When the polarizability density $N\alpha$ approaches the critical threshold $3\epsilon_0$, the denominator in the Clausius–Mossotti equation vanishes, leading to an electrostatic instability (the *polarization catastrophe*) that drives the transition into a ferroelectric state.
- **Symmetry Hierarchy:** Inversion symmetry $\mathcal{I}$ is the ultimate gatekeeper:

$$\begin{aligned} \text{Centrosymmetric (11 classes)} &\implies \text{No Piezoelectricity} \\ \text{Non-centrosymmetric (21 classes)} &\xrightarrow{\text{exclude 432}} \text{Piezoelectric (20)} \\ &\xrightarrow{\text{unique polar axis}} \text{Pyroelectric (10)} \\ &\xrightarrow{\text{reversible } \mathbf{P}_s} \text{Ferroelectric} \end{aligned}$$

- **Lattice Dynamics and Optical Resonance:** Optical properties reflect the frequency-dependent inertia of charges. At radio/microwave frequencies ($\omega < 10^{10}\,\text{rad/s}$), molecular dipoles reorient; in the infrared ($\omega \sim 10^{13}\,\text{rad/s}$), ions vibrate, yielding Reststrahlen total reflection; at optical/UV frequencies ($\omega \sim 10^{15}$–$10^{16}\,\text{rad/s}$), only valence electrons follow the oscillation. The Lyddane–Sachs–Teller relation directly links the lattice vibration frequencies (ω_{LO}/ω_{TO}) to the static and optical dielectric constants ($\epsilon(0)/\epsilon_\infty$).

4 Comprehensive Worked Problems and Step-by-Step Model Solutions

This section contains rigorous, step-by-step solutions for the benchmark problems covering Module 5B. Every numerical result has been derived from first principles and verified with analytical precision.

Problem Problem P1: Clausius–Mossotti Relation and Local Field Enhancement

A dielectric material has a dimensionless polarizability density $x \equiv \dfrac{N\alpha}{3\epsilon_0} = 0.25$.

(a) Calculate the relative permittivity (dielectric constant) ϵ_r of the material.

(b) Determine the ratio of the local microscopic electric field to the macroscopic field, E_{loc}/E.

Model Solution & Step-by-Step Analysis: Problem P1

Part (a): Calculation of ϵ_r

The Clausius–Mossotti relation connects macroscopic permittivity ϵ_r to microscopic polarizability α:

$$\frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{N\alpha}{3\epsilon_0} \equiv x \tag{69}$$

Solving this equation explicitly for ϵ_r:

$$\epsilon_r - 1 = x(\epsilon_r + 2) \implies \epsilon_r(1 - x) = 1 + 2x \implies \epsilon_r = \frac{1 + 2x}{1 - x} \tag{70}$$

Substituting the given value $x = 0.25 = 1/4$:

$$\epsilon_r = \frac{1 + 2(0.25)}{1 - 0.25} = \frac{1 + 0.50}{0.75} = \frac{1.50}{0.75} = \mathbf{2.00} \tag{71}$$

Part (b): Local Field Ratio E_{loc}/E

In a medium with cubic site symmetry, the Lorentz local field is given by:

$$\mathbf{E}_{\text{loc}} = \mathbf{E} + \frac{\mathbf{P}}{3\epsilon_0} \tag{72}$$

Using the constitutive relation $\mathbf{P} = \epsilon_0(\epsilon_r - 1)\mathbf{E}$:

$$\mathbf{E}_{\text{loc}} = \mathbf{E} + \frac{\epsilon_0(\epsilon_r - 1)\mathbf{E}}{3\epsilon_0} = \left[1 + \frac{\epsilon_r - 1}{3}\right]\mathbf{E} = \left(\frac{\epsilon_r + 2}{3}\right)\mathbf{E} \tag{73}$$

Substituting $\epsilon_r = 2$:

$$\frac{E_{\text{loc}}}{E} = \frac{2 + 2}{3} = \frac{\mathbf{4}}{\mathbf{3}} \approx 1.333 \tag{74}$$

Physical Insight: Due to the constructive dipole field from the Lorentz cavity surface bound charges, the local field experienced by an atom inside the dielectric is 33.3% larger than the macroscopic Maxwell field.

Problem Problem P2: Inversion Symmetry and Piezoelectric Point Groups

(a) Prove rigorously using Neumann's principle that any crystal possessing a centre of inversion cannot exhibit piezoelectricity.

(b) Out of the 32 crystallographic point groups, state exactly how many are non-centrosymmetric, and how many of these are piezoelectric. Name the exception.

Model Solution & Step-by-Step Analysis: Problem P2

Part (a): Symmetry Proof

The direct piezoelectric effect relates the induced electric polarization vector $\mathbf{P}$ (rank-1 polar tensor) to the applied mechanical stress tensor $\boldsymbol{\sigma}$ (rank-2 symmetric tensor):

$$P_i = d_{ijk}\sigma_{jk} \tag{75}$$

where d_{ijk} is the third-rank piezoelectric tensor. Under the spatial inversion operation $\mathcal{I} : \mathbf{r} \to -\mathbf{r}$ (transformation matrix $a_{ij} = -\delta_{ij}$):

- Polar vector $\mathbf{P}$ transforms as $P_i \to -P_i$ (odd parity).
- Second-rank stress tensor σ_{jk} transforms as $\sigma_{jk} \to (-1)(-1)\sigma_{jk} = +\sigma_{jk}$ (even parity).

According to Neumann's principle, if a crystal possesses inversion symmetry, its physical property tensor must remain invariant under $\mathcal{I}$:

$$d_{ijk} = a_{il}a_{jm}a_{kn}d_{lmn} = (-1)(-1)(-1)d_{ijk} = -d_{ijk} \tag{76}$$

The only mathematical solution to $d_{ijk} = -d_{ijk}$ is:

$$d_{ijk} \equiv 0 \quad \text{for all } i, j, k \tag{77}$$

Thus, a centrosymmetric crystal cannot exhibit piezoelectricity.

Part (b): Crystallographic Group Classification

Of the 32 crystallographic point groups:

- 11 point groups possess a centre of inversion ($C_i, C_{2h}, D_{2h}, C_{4h}, D_{4h}, S_6, D_{3d}, C_{6h}, D_{6h}, T_h, O_h$). None are piezoelectric.
- The remaining $32 - 11 = \mathbf{21}$ point groups lack an inversion centre.
- Among these 21 non-centrosymmetric groups, **20** exhibit piezoelectricity.
- The sole non-centrosymmetric exception is the cubic point group **432** (O in Schönflies notation). In group 432, the combination of multiple mutually orthogonal rotation axes produces high symmetry that forces all components of the third-rank tensor d_{ijk} to vanish identically, despite the absence of an inversion centre.

Problem Problem P3: Ferroelectric Susceptibility and the Curie–Weiss Law

A ferroelectric material in its paraelectric phase ($T > T_c$) obeys the Curie–Weiss relation:

$$\epsilon_r = \frac{C}{T - T_0} \tag{78}$$

with Curie constant $C = 1.20 \times 10^5\,\mathrm{K}$ and Curie–Weiss temperature $T_0 = 393\,\mathrm{K}$ (120°C). At what absolute temperature T does the relative permittivity drop to $\epsilon_r = 600$?

Model Solution & Step-by-Step Analysis: Problem P3

Rearranging the Curie–Weiss law to solve for temperature T:

$$T - T_0 = \frac{C}{\epsilon_r} \implies T = T_0 + \frac{C}{\epsilon_r} \tag{79}$$

Substituting the given parameters:

$$T = 393\,\mathrm{K} + \frac{1.20 \times 10^5\,\mathrm{K}}{600} \tag{80}$$

$$= 393\,\mathrm{K} + 200\,\mathrm{K} = \mathbf{593}\,\mathrm{K} \quad (320^\circ\mathrm{C}) \tag{81}$$

Check & Physical Context: At $T \to T_0^+$, $\epsilon_r \to \infty$. As temperature increases 200 K above T_0 to 593 K, thermal fluctuations progressively disorder the dipolar displacements, reducing the dielectric response to $\epsilon_r = 600$, which is still three orders of magnitude larger than ordinary non-ferroelectric insulators.

Problem Problem P4: Complex Permittivity, Refractive Index, and Reflectance

An absorbing solid has real and imaginary components of the complex dielectric function given by $\epsilon_1 = 4.0$ and $\epsilon_2 = 1.0$ at a laser frequency ω.

(a) Calculate the real refractive index n and the extinction coefficient κ.

(b) Compute the normal-incidence optical power reflectance R.

Model Solution & Step-by-Step Analysis: Problem P4

Part (a): Determination of n and κ

The complex dielectric function $\tilde{\epsilon} = \epsilon_1 + i\epsilon_2$ is related to the complex refractive index $\tilde{n} = n + i\kappa$ by:

$$\tilde{\epsilon} = \tilde{n}^2 = (n + i\kappa)^2 = (n^2 - \kappa^2) + i(2n\kappa) \tag{82}$$

Equating real and imaginary parts:

$$\epsilon_1 = n^2 - \kappa^2 = 4.0, \qquad \epsilon_2 = 2n\kappa = 1.0 \tag{83}$$

The magnitude of the dielectric function is:

$$|\tilde{\epsilon}| = \sqrt{\epsilon_1^2 + \epsilon_2^2} = \sqrt{4.0^2 + 1.0^2} = \sqrt{17} \approx 4.1231 \tag{84}$$

Since $n^2 + \kappa^2 = |\tilde{n}|^2 = |\tilde{\epsilon}|$, we solve directly for n and κ:

$$n = \sqrt{\frac{|\tilde{\epsilon}| + \epsilon_1}{2}} = \sqrt{\frac{4.1231 + 4.0}{2}} = \sqrt{\frac{8.1231}{2}} = \sqrt{4.06155} \approx \mathbf{2.0153} \tag{85}$$

$$\kappa = \sqrt{\frac{|\tilde{\epsilon}| - \epsilon_1}{2}} = \sqrt{\frac{4.1231 - 4.0}{2}} = \sqrt{\frac{0.1231}{2}} = \sqrt{0.06155} \approx \mathbf{0.2481} \tag{86}$$

Cross-check: $n^2 - \kappa^2 = 4.06155 - 0.06155 = 4.0000 = \epsilon_1$, and $2n\kappa = 2(2.0153)(0.2481) = 1.000 = \epsilon_2$.

Part (b): Normal-Incidence Power Reflectance R

From Fresnel's reflection formula at normal incidence:

$$R = \frac{(n-1)^2 + \kappa^2}{(n+1)^2 + \kappa^2} \tag{87}$$

Substituting $n = 2.0153$ and $\kappa = 0.2481$:

$$(n-1)^2 + \kappa^2 = (1.0153)^2 + (0.2481)^2 = 1.0308 + 0.06155 = 1.0924 \tag{88}$$

$$(n+1)^2 + \kappa^2 = (3.0153)^2 + (0.2481)^2 = 9.0920 + 0.06155 = 9.1536 \tag{89}$$

$$R = \frac{1.0924}{9.1536} \approx \mathbf{0.1193} \quad (\approx \mathbf{11.93}\%) \tag{90}$$

Problem Problem P5: Lyddane–Sachs–Teller Relation and Optical Phonon Splitting

In a polar crystal, the static dielectric constant is measured to be $\epsilon(0) = 1000$, and the high-frequency optical dielectric constant is $\epsilon_\infty = 5.0$. The transverse optical (TO) phonon frequency is measured by infrared spectroscopy to be $\omega_{TO} = 1.0 \times 10^{13}\,\text{rad s}^{-1}$. Find the longitudinal optical (LO) phonon frequency ω_{LO} and the ratio ω_{LO}/ω_{TO}.

Model Solution & Step-by-Step Analysis: Problem P5

The Lyddane–Sachs–Teller (LST) relation links the ratio of phonon mode frequencies to macroscopic permittivity values:

$$\frac{\omega_{LO}^2}{\omega_{TO}^2} = \frac{\epsilon(0)}{\epsilon_\infty} \tag{91}$$

Taking the square root:

$$\omega_{LO} = \omega_{TO}\sqrt{\frac{\epsilon(0)}{\epsilon_\infty}} \tag{92}$$

Substituting the given numerical values:

$$\frac{\epsilon(0)}{\epsilon_\infty} = \frac{1000}{5.0} = 200 \tag{93}$$

$$\sqrt{200} = 10\sqrt{2} \approx 14.1421 \tag{94}$$

Therefore:

$$\omega_{LO} = (1.0 \times 10^{13}\,\text{rad s}^{-1}) \times 14.1421 = \mathbf{1.4142 \times 10^{14}}\,\text{rad s}^{-1} \tag{95}$$

$$\frac{\omega_{LO}}{\omega_{TO}} = \mathbf{14.142} \tag{96}$$

Physical Meaning: In a longitudinal mode, the macroscopic polarization creates a macroscopic restoring depolarizing field ($\mathbf{E} = -\mathbf{P}/\epsilon_0$), stiffening the effective spring constant and dramatically shifting ω_{LO} upward by a factor of ≈ 14.1 into the near infrared. In contrast, in a transverse mode, $\nabla \cdot \mathbf{P} = 0$, so no macroscopic depolarizing field arises.

Problem Problem P6: Separating Electronic and Ionic Polarizability: NaCl vs. Diamond

For rocksalt (NaCl), the static permittivity is $\epsilon_r(0) = 5.62$ and the optical refractive index is $n = 1.50$. For diamond, $\epsilon_r(0) = 5.68$ and $n = 2.38$.

(a) Compute the Clausius–Mossotti factor $Y \equiv \dfrac{\epsilon_r - 1}{\epsilon_r + 2}$ for both materials under static and optical conditions (recalling $\epsilon_r(\text{opt}) = n^2$).

(b) Determine what fraction of the static dielectric response of NaCl is of ionic origin.

(c) Explain physically why the two values of Y coincide exactly for diamond, but diverge dramatically for NaCl.

Model Solution & Step-by-Step Analysis: Problem P6

Part (a): Computation of Y
For **NaCl**:

$$\text{Static: } \epsilon_r(0) = 5.62 \implies Y^{(\text{static})} = \frac{5.62-1}{5.62+2} = \frac{4.62}{7.62} \approx \mathbf{0.6063} \tag{97}$$

$$\text{Optical: } \epsilon_r(\text{opt}) = n^2 = 1.50^2 = 2.25 \implies Y^{(n)} = \frac{2.25-1}{2.25+2} = \frac{1.25}{4.25} \approx \mathbf{0.2941} \tag{98}$$

For **Diamond**:

$$\text{Static: } \epsilon_r(0) = 5.68 \implies Y^{(\text{static})} = \frac{5.68-1}{5.68+2} = \frac{4.68}{7.68} \approx \mathbf{0.6094} \tag{99}$$

$$\text{Optical: } \epsilon_r(\text{opt}) = n^2 = 2.38^2 = 5.6644 \implies Y^{(n)} = \frac{5.6644-1}{5.6644+2} = \frac{4.6644}{7.6644} \approx \mathbf{0.6086} \tag{100}$$

Part (b): Ionic Share in NaCl
In the optical regime ($\omega \sim 10^{15}\,\text{rad s}^{-1}$), heavy ions cannot follow the electric field; only electronic polarizability contributes:

$$Y^{(n)} = \frac{N\alpha_e}{3\epsilon_0} = 0.2941, \qquad Y^{(\text{static})} = \frac{N(\alpha_e+\alpha_i)}{3\epsilon_0} = 0.6063 \tag{101}$$

The ionic contribution to Y is:

$$Y_{\text{ionic}} = Y^{(\text{static})} - Y^{(n)} = 0.6063 - 0.2941 = \mathbf{0.3122} \tag{102}$$

Expressed as a fraction of the total static susceptibility contribution:

$$f_{\text{ionic}} = 1 - \frac{n^2-1}{\epsilon_r(0)-1} = 1 - \frac{2.25-1}{5.62-1} = 1 - \frac{1.25}{4.62} = 1 - 0.2706 \approx 72.9\% \tag{103}$$

Or evaluated directly from the Clausius–Mossotti polarizability shares:

$$\frac{\alpha_i}{\alpha_e+\alpha_i} = \frac{0.3122}{0.6063} \approx \mathbf{51.5}\% \tag{104}$$

In either convention, the ionic displacement contributes more than half of the static polarization.

Part (c): Physical Reason for the Contrast

- **Diamond:** Carbon atoms are held together by purely homopolar, covalent bonds with identical electronegativity. No net ionic charges exist on lattice sites ($\pm q = 0$). Hence, relative displacement of adjacent carbon sublattices produces zero net electric dipole moment ($\alpha_i = 0$). The only polarization mechanism available is the distortion of the electronic cloud, so $\alpha_{\text{static}} = \alpha_{\text{optical}} = \alpha_e$, yielding $Y^{(\text{static})} \approx Y^{(n)} = 0.609$.
- **NaCl:** The lattice is composed of alternating positive Na^+ and negative Cl^- ions. In a static electric field, the entire positive and negative sublattices displace in opposite directions, producing a massive ionic dipole moment that accounts for over half the dielectric response. In the visible spectrum, the field alternates too rapidly ($10^{15}\,\text{Hz}$) for the heavy nuclei to respond, freezing out the ionic contribution and leaving only α_e.

Problem Problem P7: Depolarization Fields in Spherical vs. Needle Geometries

A dielectric specimen is uniformly polarized with $\mathbf{P} = 0.10\,\mathrm{C\ m^{-2}}\hat{\mathbf{z}}$.

(a) Calculate the internal depolarizing field $\mathbf{E}_{\mathrm{dep}}$ inside a spherical specimen.

(b) Calculate $\mathbf{E}_{\mathrm{dep}}$ inside a long, slender needle oriented parallel to $\mathbf{P}$.

(c) In which of the two geometries is the local field $\mathbf{E}_{\mathrm{loc}}$ at an atomic site closest to the externally applied field $\mathbf{E}_0$?

Model Solution & Step-by-Step Analysis: Problem P7

Part (a): Spherical Specimen

For a sphere, the depolarization factor along any principal axis is $N_x = N_y = N_z = 1/3$. The depolarizing field is:

$$\mathbf{E}_{\mathrm{dep}} = -\frac{N_z P}{\epsilon_0}\hat{\mathbf{z}} = -\frac{P}{3\epsilon_0}\hat{\mathbf{z}} \tag{105}$$

Using $\epsilon_0 = 8.85419 \times 10^{-12}\,\mathrm{F\ m^{-1}}$:

$$\mathbf{E}_{\mathrm{dep}} = -\frac{0.10\,\mathrm{C\ m^{-2}}}{3 \times 8.85419 \times 10^{-12}\,\mathrm{F\ m^{-1}}}\hat{\mathbf{z}} = \mathbf{-3.765 \times 10^{9}\,V\ m^{-1}\,\hat{z}} \tag{106}$$

This is a colossal opposing electric field on the order of $3.765\,\mathrm{GV\ m^{-1}}$.

Part (b): Needle Specimen Parallel to P

For a long cylinder or needle of length $L \gg R$ aligned along $\hat{\mathbf{z}}$, the bound surface charges occur only on the distant flat end-caps:

$$N_z \to 0, \qquad N_x = N_y = \frac{1}{2} \tag{107}$$

Therefore, along the needle axis:

$$\mathbf{E}_{\mathrm{dep}} = -N_z\frac{\mathbf{P}}{\epsilon_0} = \mathbf{0} \tag{108}$$

Part (c): Comparison of Local Fields

The macroscopic field inside the specimen is $\mathbf{E} = \mathbf{E}_0 + \mathbf{E}_{\mathrm{dep}}$. The local field at an atomic site with cubic symmetry is:

$$\mathbf{E}_{\mathrm{loc}} = \mathbf{E} + \frac{\mathbf{P}}{3\epsilon_0} = \mathbf{E}_0 + \mathbf{E}_{\mathrm{dep}} + \frac{\mathbf{P}}{3\epsilon_0} = \mathbf{E}_0 + \left(\frac{1}{3} - N_z\right)\frac{\mathbf{P}}{\epsilon_0} \tag{109}$$

- **For the Sphere ($N_z = 1/3$):**

$$\mathbf{E}_{\mathrm{loc}} = \mathbf{E}_0 + \left(\frac{1}{3} - \frac{1}{3}\right)\frac{\mathbf{P}}{\epsilon_0} = \mathbf{E}_0 \tag{110}$$

  Remarkably, inside a sphere, the macroscopic depolarizing field $\mathbf{E}_{\mathrm{dep}} = -\mathbf{P}/3\epsilon_0$ *exactly cancels* the Lorentz cavity field $\mathbf{E}_2 = +\mathbf{P}/3\epsilon_0$. Consequently, $\mathbf{E}_{\mathrm{loc}}$ equals the applied external field $\mathbf{E}_0$!

- **For the Needle ($N_z = 0$):**

$$\mathbf{E}_{\mathrm{loc}} = \mathbf{E}_0 + \frac{\mathbf{P}}{3\epsilon_0} = \mathbf{E} + \frac{\mathbf{P}}{3\epsilon_0} \tag{111}$$

  Here $\mathbf{E}_{\mathrm{loc}}$ exceeds the macroscopic field by $+3.765 \times 10^{9}\,\mathrm{V\ m^{-1}}$.

Conclusion: In terms of matching the external applied field $\mathbf{E}_0$, the local field inside a **spherical** specimen is identical to $\mathbf{E}_0$; whereas inside a needle, the macroscopic field matches $\mathbf{E}_0$ while the local field is enhanced by $\mathbf{P}/3\epsilon_0$.

5 Oral Conceptual Examination Questions and Rigorous Model Answers

These four foundational questions test physical reasoning and are frequent topics in postgraduate comprehensive vivas and doctoral qualifying examinations.

Conceptual Checkpoint: Viva Question 1: High-Frequency Limit of Gaseous Permittivity

Question: Why does the relative permittivity of a gas approach unity ($\epsilon_r \to 1$) at very high frequencies (e.g., ultraviolet and X-ray regimes)?
Comprehensive Model Answer:

1. **Driven Oscillator Dynamics:** The displacement $\mathbf{x}(t)$ of a bound electron with natural resonance frequency ω_0, damping γ, and mass m in an alternating electric field $\mathbf{E}(t) = \mathbf{E}_0 e^{-i\omega t}$ is governed by the equation of motion:

$$m\ddot{\mathbf{x}} + m\gamma\dot{\mathbf{x}} + m\omega_0^2\mathbf{x} = -e\mathbf{E}_0 e^{-i\omega t}$$

   The steady-state solution gives the frequency-dependent polarizability:

$$\alpha(\omega) = \frac{e^2/m}{\omega_0^2 - \omega^2 - i\gamma\omega}$$

2. **Asymptotic Behavior at $\omega \gg \omega_0$:** When the driving frequency far exceeds the binding resonant frequency of the electrons ($\omega \gg \omega_0, \gamma$), the inertia of the electron prevents it from following the field. The restoring and damping forces become negligible compared to the acceleration term $m\ddot{\mathbf{x}} \sim -m\omega^2\mathbf{x}$. The polarizability scales as:

$$\alpha(\omega) \approx -\frac{e^2}{m\omega^2} \xrightarrow{\omega\to\infty} 0$$

3. **Permittivity Limit:** In a dilute gas, $\epsilon_r(\omega) = 1 + \dfrac{N\alpha(\omega)}{\epsilon_0} \approx 1 - \dfrac{Ne^2}{\epsilon_0 m\omega^2} = 1 - \frac{\omega_p^2}{\omega^2}$. As $\omega \to \infty$, the second term vanishes asymptotically, and $\epsilon_r \to \mathbf{1}$. Physically, at X-ray frequencies, the electromagnetic field oscillates so rapidly that the bound electrons cannot respond, and the medium behaves electrodynamically as free vacuum.

Conceptual Checkpoint: Viva Question 2: Pyroelectricity vs. Spontaneous Polarization in Crystals

Question: Can a non-centrosymmetric crystal exhibit piezoelectricity without being pyroelectric? Can it be pyroelectric without being ferroelectric? Explain the symmetry hierarchy.
Comprehensive Model Answer:

1. **Piezoelectricity vs. Pyroelectricity:** Yes. Out of the 32 crystal point groups, 21 lack an inversion centre, and 20 are piezoelectric. However, piezoelectricity only requires an induced dipole under *external mechanical stress* ($P_i = d_{ijk}\sigma_{jk}$). In contrast, **pyroelectricity** requires a permanent, spontaneous electric dipole moment $\mathbf{P}_s$ in the absence of stress or applied field. Under crystal symmetry transformations, a permanent vector $\mathbf{P}_s$ must remain invariant. This is only possible if the point group possesses a *unique polar rotation axis* that is not inverted or flipped by perpendicular two-fold axes or mirror planes. Only **10 point groups** satisfy this condition: $1, 2, m, mm2, 4, 4mm, 3, 3m, 6, 6mm$ (all polar groups). The remaining 10 piezoelectric groups (such as quartz, point group 32, or cubic $\bar{4}3m$) lack a unique polar axis; their internal dipoles cancel out symmetrically in

the ground state.

2. **Pyroelectricity vs. Ferroelectricity:** A crystal can be pyroelectric without being ferroelectric! A **ferroelectric** is a subgroup of pyroelectric crystals in which the spontaneous polarization $\mathbf{P}_s$ can be *reoriented or reversed between two or more stable orientations by an achievable external electric field*. Certain pyroelectric crystals (such as wurtzite ZnO or tourmaline) possess a permanent $\mathbf{P}_s$, but their crystal lattices are so rigid that the electric field required to reverse $\mathbf{P}_s$ exceeds the dielectric breakdown strength of the crystal ($E_{\text{switch}} > E_{\text{breakdown}}$). Thus, they are pyroelectric, but not ferroelectric.

Conceptual Checkpoint: Viva Question 3: Imaginary Refractive Index in the Reststrahlen Band

Question: Inside the Reststrahlen band between ω_{TO} and ω_{LO}, why is the refractive index purely imaginary in the absence of damping, and what is the physical consequence for reflection?

Comprehensive Model Answer:

1. **Dielectric Function in the Phonon Gap:** For an undamped polar lattice ($\gamma \to 0$), the Lyddane–Sachs–Teller dielectric function is:

$$\epsilon(\omega) = \epsilon_\infty + \frac{\epsilon(0) - \epsilon_\infty}{1 - (\omega/\omega_{TO})^2} = \epsilon_\infty \left(\frac{\omega_{LO}^2 - \omega^2}{\omega_{TO}^2 - \omega^2} \right)$$

   For frequencies between the transverse and longitudinal optical phonons, $\omega_{TO} < \omega < \omega_{LO}$:

$$\omega^2 - \omega_{TO}^2 > 0 \quad \text{and} \quad \omega_{LO}^2 - \omega^2 > 0 \implies \boldsymbol{\epsilon(\omega) < 0}$$

   The dielectric function is strictly **negative and real**.

2. **Refractive Index Calculation:** The complex refractive index $\tilde{n} = n + i\kappa = \sqrt{\epsilon(\omega)}$. Since $\epsilon(\omega) = -|\epsilon|$, we have:

$$\tilde{n} = \sqrt{-|\epsilon|} = i\sqrt{|\epsilon|} \implies \boldsymbol{n = 0, \quad \kappa = \sqrt{|\epsilon|}}$$

   The real refractive index n is identically zero, and $\tilde{n}$ is purely imaginary.

3. **Physical Consequence — Total Reflection:** The electromagnetic wavevector in the medium is $k = \tilde{n}\omega/c = i\kappa\omega/c$. The spatial wave solution becomes an evanescent, exponentially decaying wave:

$$\mathbf{E}(z,t) = \mathbf{E}_0 e^{-(\kappa\omega/c)z} e^{-i\omega t}$$

   No propagating electromagnetic energy can penetrate the bulk crystal. Substituting $n = 0$ into Fresnel's normal-incidence reflectance:

$$R = \frac{(n-1)^2 + \kappa^2}{(n+1)^2 + \kappa^2} = \frac{(-1)^2 + \kappa^2}{(+1)^2 + \kappa^2} = \frac{1 + \kappa^2}{1 + \kappa^2} = \mathbf{1.00} \quad (\mathbf{100}\%)$$

   The crystal acts as an ideal optical mirror, exhibiting total reflection for all infrared frequencies in the Reststrahlen band.

Conceptual Checkpoint: Viva Question 4: Lorentz–Lorenz vs. Static Clausius–Mossotti Equation

Question: Why does the Lorentz–Lorenz relation determine the *electronic polarizability* alone, while the static Clausius–Mossotti relation yields the *sum of electronic and ionic polarizabilities*?

Comprehensive Model Answer:

1. **Mathematical Formulation:** Both relations stem from the same cavity electrostatics $\mathbf{E}_{\text{loc}} = \mathbf{E} + \mathbf{P}/3\epsilon_0$:

$$\text{Static Clausius–Mossotti: } \frac{\epsilon_r(0) - 1}{\epsilon_r(0) + 2} = \frac{N\alpha_{\text{total}}}{3\epsilon_0} = \frac{N(\alpha_e + \alpha_i + \alpha_{\text{dip}})}{3\epsilon_0}$$

$$\text{Optical Lorentz–Lorenz: } \frac{n^2-1}{n^2+2} = \frac{N\alpha_e}{3\epsilon_0} \quad (\text{using } \epsilon_r(\omega_{\text{opt}}) = n^2)$$

2. **Microscopic Inertial Separation:** The distinction is entirely governed by the characteristic dynamical timescales and resonant frequencies of the physical charge carriers:
   - **Permanent Dipolar Reorientation (α_{dip}):** Governed by molecular rotational diffusion. In liquids and solids, dipolar relaxation times are $\tau \sim 10^{-10}$–$10^{-9}\,\text{s}$ ($\omega \sim 10^{9}$–$10^{10}\,\text{rad s}^{-1}$). Above microwave frequencies, molecular dipoles cannot rotate fast enough to keep up with the alternating field.
   - **Ionic Displacement (α_i):** Involves the physical displacement of whole atomic nuclei ($M_{\text{ion}} \sim 10^4 m_e$). The optical phonon resonant frequency is $\omega_{TO} \sim \sqrt{K/M_{\text{ion}}} \sim 10^{13}\,\text{rad s}^{-1}$ (infrared spectrum).
   - **Electronic Cloud Distortion (α_e):** Involves the displacement of electrons with minuscule mass m_e. The electronic resonant frequencies lie in the ultraviolet $\omega_0 \sim \sqrt{K/m_e} \sim 10^{16}\,\text{rad s}^{-1}$.
3. **Diagnostic Power:** In the visible spectrum ($\omega \sim 3 \times 10^{15}\,\text{rad s}^{-1}$), the frequency is far above the ionic resonance ($\omega \gg \omega_{TO}$), completely freezing out ionic displacements ($\alpha_i \to 0$), but well below electronic resonance ($\omega \ll \omega_0$). Consequently, the optical refractive index n measures α_e in isolation. Subtracting the Lorentz–Lorenz value from the static Clausius–Mossotti value provides an unambiguous experimental tool to decouple electronic and ionic polarizabilities.